

The Cyclic Causal-State Hypothesis

A Speculative Lorentz-Scalar Extension of Relativistic Causal Kinematics

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September 2026

Status: speculative theoretical framework - not established physics

Abstract

This paper develops a speculative extension of relativistic kinematics motivated by the familiar special-relativistic relation $d\tau/dt = \sqrt{1 - v^2/c^2}$. The standard theory fixes the future-directed timelike branch and does not permit a massive particle to cross the light cone at finite energy. We first show why a naive signed continuation fails: fixed nonzero invariant mass forces the Lorentz factor, energy, and momentum to diverge as $|v|$ approaches c , while future- and past-directed timelike mass shells are disconnected. We then introduce an internal Lorentz-scalar state φ that changes the Minkowski norm of four-momentum according to $P \cdot P = m_0^2 c^2 \cos(2\varphi)$. A finite auxiliary parameterization passes continuously through timelike, null, spacelike, null, and oppositely oriented timelike sectors. A constrained worldline action and a scalar-field extension are proposed as minimal toy dynamics. Conservation, causality restrictions, the low-energy special-relativistic limit, and falsifiable signatures are discussed. The construction is a hypothesis: no experimental evidence is claimed for causal-sector transitions, superluminal propagation, or backward-time transport.

1. Motivation and Boundary of the Claim

For a future-directed timelike worldline in special relativity, proper time obeys

$$d\tau/dt = \sqrt{1 - v^2/c^2}, \quad |v| < c.$$

Squaring this expression suggests a geometric relation between a temporal rate and spatial velocity, but the square root in standard SR is the positive branch for an oriented future-directed worldline. A negative temporal branch is therefore not derived from SR. It is an additional hypothesis.

The purpose of this paper is to determine what mathematical structure would be required if one nevertheless postulated a continuous transition from a future-directed timelike state to an oppositely oriented timelike state. The analysis deliberately separates established SR identities from new postulates.

2. The Original Orientation-State Circle

Define the auxiliary temporal-rate variable

$$T \equiv c \, d\tau/dt.$$

After squaring the standard timelike relation,

$$T^2 + v^2 = c^2.$$

A signed extension can be parameterized by an angle θ :

$$v = c \sin \theta, \quad T = c \cos \theta.$$

This produces the intuitive cycle $(v, T) = (0, +c) \rightarrow (+c, 0) \rightarrow (0, -c) \rightarrow (-c, 0) \rightarrow (0, +c)$. However, the lower semicircle is a branch choice introduced by the model. Moreover, this circle is an auxiliary Euclidean state-space picture, not the geometry of Minkowski spacetime.

3. Why the Naive Fixed-Mass Continuation Fails

With $\beta = v/c = \sin \theta$, a literal signed continuation gives

$$\gamma_{\text{signed}} = \sec \theta, \quad E = m c^2 \sec \theta, \quad p = m c \tan \theta.$$

At $\theta = \pi/2$ and $3\pi/2$ these expressions diverge. The invariant nevertheless remains algebraically intact:

$$E^2 - p^2 c^2 = m^2 c^4 (\sec^2 \theta - \tan^2 \theta) = m^2 c^4.$$

The divergence is not an arbitrary inconvenience. For fixed $m > 0$, the mass shell

$$E^2 - p^2 c^2 = m^2 c^4$$

has two disconnected timelike sheets, $E > 0$ and $E < 0$. A continuous finite path cannot move from one sheet to the other while remaining on the same nonzero-mass timelike shell. In rapidity language, $\beta = \tanh \eta$ and $\gamma = \cosh \eta$; the light boundary corresponds to $|\eta| \rightarrow \infty$. Thus the pole at c is not removed by a simple change of ordinary timelike coordinates.

4. Finite Energy-Momentum Transition Parameterization

To investigate a finite path, introduce a new angular state φ and a fixed scale E_0 :

$$E = E_0 \cos \varphi, \quad pc = E_0 \sin \varphi.$$

The auxiliary Euclidean norm is finite and conserved along the parameterized circle:

$$E^2 + p^2 c^2 = E_0^2.$$

The physically relevant Minkowski norm is instead

$$E^2 - p^2 c^2 = E_0^2 \cos(2\varphi).$$

Writing $m_0 = E_0/c^2$ motivates the central hypothesis

$$P^u P_u = m_0^2 c^2 \cos(2\varphi).$$

φ	E	pc	PP	Causal character
0	$+E_0$	0	$+m_0^2 c^2$	Future timelike
$\pi/4$	$+E_0/\sqrt{2}$	$+E_0/\sqrt{2}$	0	Null
$\pi/2$	0	$+E_0$	$-m_0^2 c^2$	Spacelike
$3\pi/4$	$-E_0/\sqrt{2}$	$+E_0/\sqrt{2}$	0	Null
π	$-E_0$	0	$+m_0^2 c^2$	Opposite timelike orientation

If the usual relation $v = pc^2/E$ is used as a derived coordinate, then

$$v = c \tan \varphi.$$

Consequently $v = c$ at $\varphi = \pi/4$, diverges as E passes through zero at $\varphi = \pi/2$, and equals $-c$ at $\varphi = 3\pi/4$. The underlying pair (E, p) remains finite. Thus v is not a globally valid coordinate for this hypothesized transition.

5. Matching Spacetime and Momentum Causal Character

A useful 1+1-dimensional auxiliary trajectory is obtained by defining

$$dt/d\lambda = \cos \varphi, \quad dx/d\lambda = c \sin \varphi.$$

Then

$$dx/dt = c \tan \varphi$$

and the tangent norm becomes

$$(ds/d\lambda)^2 = c^2 \cos^2\varphi - c^2 \sin^2\varphi = c^2 \cos(2\varphi).$$

The same $\cos(2\varphi)$ therefore controls both the spacetime tangent norm and the four-momentum norm. In the timelike sector, ordinary SR time dilation is recovered:

$$d\tau/dt = \sqrt{(\cos 2\varphi)/\cos \varphi} = \sqrt{(1 - v^2/c^2)}.$$

This equality is a consistency check on the timelike portion of the toy parameterization, not evidence that the spacelike continuation is realized in nature.

6. A Simple Auxiliary Evolution Law

For illustration, take

$$d\varphi/d\lambda = \omega, \quad \varphi = \omega\lambda + \varphi_0.$$

Then

$$\begin{aligned} E(\lambda) &= E_0 \cos(\omega\lambda + \varphi_0), & p(\lambda) &= (E_0/c) \sin(\omega\lambda + \varphi_0), \\ dE/d\lambda &= -\omega c p, & dp/d\lambda &= (\omega/c)E. \end{aligned}$$

Hence

$$d^2E/d\lambda^2 + \omega^2 E = 0, \quad d^2p/d\lambda^2 + \omega^2 p = 0.$$

For $\varphi_0 = 0$, integration of the spacetime tangent gives

$$\begin{aligned} t(\lambda) &= t_0 + \sin(\omega\lambda)/\omega, \\ x(\lambda) &= x_0 + (c/\omega)[1 - \cos(\omega\lambda)]. \end{aligned}$$

Eliminating λ yields the auxiliary circular trajectory

$$[\omega(t - t_0)]^2 + [1 - \omega(x - x_0)/c]^2 = 1.$$

At $\varphi = \pi$ the parameterized path returns to $t = t_0$ while reaching $x = x_0 + 2c/\omega$. This is a prediction of this particular toy parameterization only. It must not be confused with an experimentally established time-travel trajectory.

7. Minimal Constrained Worldline Dynamics

A more defensible formulation makes φ a dynamical Lorentz scalar rather than imposing constant angular motion. Consider the first-order worldline action

$$S = \int d\lambda [P_u \dot{X}^u + \Pi\varphi \dot{\varphi} - (N/2)(P^u P_u - m_0^2 c^2 \cos 2\varphi) - \Pi\varphi^2/(2I) - V(\varphi)].$$

Variation with respect to N enforces the causal mass-shell constraint:

$$P^u P_u = m_0^2 c^2 \cos 2\varphi.$$

Variation with respect to P_u gives

$$\dot{X}^u = N P^u.$$

The internal canonical relation is

$$\dot{\varphi} = \Pi\varphi/I.$$

The φ equation has the generic nonlinear form

$$I \ddot{\varphi} + N m_0^2 c^2 \sin(2\varphi) + V'(\varphi) = 0,$$

up to sign conventions associated with the precise first-order action and parameter gauge. The point is that φ is governed dynamically; constant $\dot{\varphi}$ is at most a special solution.

8. Transition Potential and Ordinary-Matter Stability

A simple periodic toy potential is

$$V(\varphi) = V_0 \sin^2 \varphi = (V_0/2)[1 - \cos(2\varphi)].$$

The ordinary state $\varphi = 0$ is a minimum. The first null boundary at $\varphi = \pi/4$ lies at $V = V_0/2$, while $\varphi = \pi/2$ lies at $V = V_0$. An internal mechanical analogue therefore has

$$\mathcal{E}\varphi = (I/2) \dot{\varphi}^2 + V_{\text{eff}}(\varphi).$$

A transition requires sufficient excitation to leave the low-energy basin. This is a model-building device, not a known physical mechanism.

9. Scalar-Field Extension

The internal state can be promoted from $\varphi(\lambda)$ to a Lorentz-scalar field $\varphi(x)$. A minimal field sector is

$$\mathcal{L}\varphi = (f^2/2) \partial_\mu \varphi \partial^\mu \varphi - V(\varphi),$$

where f supplies the normalization scale because φ is dimensionless. Its field equation is

$$f^2 \square \varphi + dV/d\varphi = 0.$$

For $V = V_0 \sin^2 \varphi$, the nonlinear term is proportional to $\sin(2\varphi)$, producing a sine-Gordon-like structure. A complete theory would additionally require a consistent coupling between this field and matter. Merely writing a scalar sector does not demonstrate that causal-sector-changing excitations are stable.

10. Conservation Laws

Changing the invariant norm of a particle four-momentum cannot occur for an isolated free particle while that same four-momentum remains conserved. The transition must therefore be an interaction.

$$P^\mu_{\text{total}} = P^\mu_{\text{matter}} + P^\mu_{\varphi}, \quad dP^\mu_{\text{total}}/d\lambda = 0.$$

At field-theory level the required conservation statement is

$$\partial_\mu (T^\mu \nu_{\text{matter}} + T^\mu \nu_{\varphi}) = 0.$$

For a canonical scalar field,

$$T^\mu \nu_{\varphi} = f^2 \partial^\mu \varphi \partial^\nu \varphi - g^\mu \nu [(f^2/2) \partial_\alpha \varphi \partial^\alpha \varphi - V(\varphi)].$$

Thus the matter subsystem may exchange energy and momentum with the transition field while total stress-energy remains conserved.

11. Causality and the FTL Signalling Problem

Spacelike propagation is the most serious conceptual obstacle. In Lorentz-invariant SR, the temporal ordering of spacelike-separated events is frame-dependent, and controllable superluminal signalling can be combined into closed causal loops. The present hypothesis therefore cannot simply declare an arbitrary FTL communication channel.

A minimal causal restriction is proposed: the spacelike sector is an intermediate state determined by the equations of motion and cannot be independently targeted or used to send freely chosen information between arbitrary spacelike-separated events. The auxiliary parameter λ supplies an ordering of states even when coordinate time t is non-monotonic.

This restriction is a postulate, not a proof of causal consistency. A complete theory must demonstrate that its equations, interactions, and allowed initial data cannot construct an antitelephone or a closed causal curve. If that cannot be shown, the model fails as a causal physical theory.

12. Energy Sign and Temporal Orientation

The finite circle uses signed E , becoming negative on the opposite timelike branch. Treating that sign as ordinary measurable negative energy would threaten stability. A safer bookkeeping interpretation separates energy magnitude from orientation:

$$E_{\text{obs}} = |E| \geq 0, \quad \sigma_t = \text{sgn}(E).$$

Here σ_t labels the orientation of the four-momentum branch. Whether such a branch can be reinterpreted consistently in a quantum theory - for example through particle/antiparticle structure - is left open. The present paper does not identify negative signed E with experimentally accessible negative-energy matter.

13. Low-Energy Correspondence with Special Relativity

For $|\varphi| \ll 1$,

$$\cos(2\varphi) \approx 1 - 2\varphi^2.$$

Hence

$$m_{\text{eff}}^2 \approx m_0^2(1 - 2\varphi^2), \quad m_{\text{eff}} \approx m_0(1 - \varphi^2).$$

As $\varphi \rightarrow 0$,

$$E^2 - p^2c^2 \rightarrow m_0^2c^4.$$

Ordinary special relativity is therefore recovered in the unexcited-state limit. Any viable version of the model must make departures sufficiently suppressed in regimes where SR has been precisely tested.

14. Candidate Observable Signatures and Falsifiability

The framework becomes physically distinguishable from a reparameterization of SR only if φ produces observable effects. Candidate low-amplitude signatures include:

- An excitation- or environment-dependent invariant-mass shift, $\Delta m/m_0 \approx -\varphi^2$ for small φ .
- A modified dispersion relation, $E^2 = p^2c^2 + m_0^2c^4 \cos(2\varphi)$, with leading correction $\Delta(E^2) \approx -2m_0^2c^4\varphi^2$.
- An additional scalar excitation associated with fluctuations of φ around its vacuum.
- Threshold behavior if interactions can excite φ toward the null boundary.

Null experimental results constrain the coupling, field amplitude, and transition scale toward a decoupling limit. In the limit $\varphi \rightarrow 0$ or vanishing matter coupling, the proposal becomes observationally indistinguishable from SR and loses independent predictive content.

15. What the Model Does Not Yet Solve

- No microscopic interaction is presently known that produces the required causal-sector transition.
- The stability and quantization of the spacelike sector have not been established.
- Negative mass-squared in field theory often signals an unstable vacuum rather than a propagating faster-than-light particle.
- A full Lorentz-covariant matter coupling and Hamiltonian analysis are still required.
- The no-independent-signalling rule must ultimately emerge from dynamics, not remain an ad hoc restriction.
- No experimental evidence currently supports the proposed internal state φ or the transition mechanism.

16. Formal Postulates of the Cyclic Causal-State Hypothesis

1. Internal-State Postulate. A relativistic excitation may possess a dimensionless Lorentz-scalar internal state φ .

2. Causal-Mass Postulate. Its effective four-momentum norm is $P \cdot P = m_0^2c^2 \cos(2\varphi)$.

3. Correspondence Postulate. The limit $\varphi \rightarrow 0$ reproduces the ordinary positive-mass special-relativistic shell.

4. Dynamical Postulate. φ evolves from an action and cannot be assigned arbitrarily.

5. Conservation Postulate. Matter and the φ sector may exchange four-momentum, but total stress-energy is conserved.

6. Causality Restriction. Intermediate spacelike states are not independently controllable arbitrary FTL signalling channels.

7. Energy-Orientation Convention. Observable energy is nonnegative; the sign used in the auxiliary circle labels temporal orientation unless a deeper theory supplies another interpretation.

17. Relationship to Standard Minkowski Geometry

Lorentz boosts are hyperbolic rotations. For ordinary future-directed timelike motion one may write

$$\beta = \tanh \eta, \quad \gamma = \cosh \eta, \quad \gamma\beta = \sinh \eta.$$

The unit timelike hyperboloid has disconnected future and past sheets. The Euclidean-looking circles in (v,T) or (E,pc) used here are therefore auxiliary compact state-space constructions, not replacements for the Minkowski metric. The Minkowski invariant remains the quantity used to classify causal character.

18. Conclusion

The investigation began with a signed geometric continuation of the time-dilation factor and exposed a fundamental obstruction: a fixed-mass timelike state cannot continuously cross the light cone at finite energy. Recasting the problem in four-momentum space suggested a more coherent speculative alternative. An internal Lorentz-scalar state φ can be postulated to vary the causal invariant continuously through timelike, null, spacelike, null, and oppositely oriented timelike sectors. A finite parameterization, worldline constraint, candidate scalar dynamics, conservation framework, low-energy SR limit, and explicit falsifiability conditions can then be written down.

The result should be understood as a structured hypothesis rather than a demonstrated extension of relativity. Its decisive unresolved tasks are a stable quantum-consistent interaction, rigorous causal consistency, and experimental evidence. Those requirements provide clear ways for the hypothesis to be developed, constrained, or rejected.

References

1. Einstein, A. (1905). On the Electrodynamics of Moving Bodies. *Annalen der Physik* 17, 891-921.
2. Minkowski, H. (1909). Space and Time. *Physikalische Zeitschrift* 10, 104-111.
3. Taylor, E. F., & Wheeler, J. A. *Spacetime Physics*. W. H. Freeman.
4. Rindler, W. *Introduction to Special Relativity*. Oxford University Press.
5. Misner, C. W., Thorne, K. S., & Wheeler, J. A. *Gravitation*. W. H. Freeman.
6. Peskin, M. E., & Schroeder, D. V. *An Introduction to Quantum Field Theory*. Westview Press.